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IPv4 HEADER

4.1 IPv4 Header

VER (4 bits)	HL (4 bits)	Services (8 bits)	Total Length (16 bits)
Identification number (16 bits)		Flags (3 bits)	Fragment offset (13 bits)
Time to Live (8 bits)		Protocol (8 bits)	Header checksum (16 bits)
Source IP Address (32 bits)			
Destination IP Address (32 bits)			
Option (0-40 bytes)			

4.1.1 Version (4 Bit)

It is used to indicate IPv4 or IPv6.

4.1.2 Header Length(4 Bit)

Header length is a 4 bit field that contains the length of header.

Minimum Header size is 20 byte.

Maximum Header size is 60 byte.

4.2 Services [8 bit]

In this Interpretation the first 3 bit are called precedence bit (Priority bit) and Next 4 bit are called types of services bits and last bit is Not used.



4.2.1 Priority

It is a 3-bit subfield ranging from 0 to 7 (000 to 111 in binary). Priority field is needed if a router is congested and need to discard some datagram, those datagrams which have the lowest priority are discarded first.

4.2.2 Types of Services

It is a 4 bit subfield. Each bit having a special meaning, although a bit can be 0 or 1. One and only one of the bits can have the value 1 in each datagram.

4.3 Total length (16 bits)

Total length = Data + Header

4.4 Identification Number (16 bits)

1. Each datagram is associated with a sequence number is called as datagram number or identification number.
2. It is used to identify all the fragment of same datagram.
3. All the fragment of same datagram will have the same identification number.

4.5 Flags

It is the 3 bit field shown in the figure.

X	D F	M F
Not Used	Don't Fragment	More Fragment

4.6 Fragment offset (13 bits)

Fragment offset indicate no of data byte ahead of this fragment in that particular packet.

4.7 TTL (8 bits)

1. TTL is used to avoid infinite looping.
2. TTL field is used to control the maximum number of hops visited by datagram.
3. When a source host sends a datagram, it stores a number in this field. Each router that process the datagram decrements this number by one. If TTL field reaches zero before the datagram arrives at its destination, then the datagram is discarded and an ICMP message is sent back to sender.



4.8 Protocol (8 bits)

1. This 8 bits field tell us which protocol is encapsulated in the IP packet.
2. At the time of traffic, some packet must be discarded. In this case it will be advantageous to know which protocol data it contains.
3. The order in which router eliminate the datagram from buffer is-
ICMP>IGMP>UDP>TCP
(01) (02) (17) (06)

4.9 Header Checksum

1. It is calculated only for header part not the data because rest of the component in packet already covered by TCP checksum.
2. Header checksum is calculated at each and every Router because IP Header might be change when packet is moving from one router to another.
3. Every router makes one modification i.e. TTL so Header checksum is calculated at every Router.
4. Fragment offset, MF, Total length, option all may be changed at a Router.

4.10 Source Address (32 bits)

This 32 bit defines the IPv4 address of source. This field remain unchanged during the time the IPv4 datagram travel from the source Host to destination Host.

4.11 Destination Address (32 bits)

This 32 bits Field defines the IPv4 address of the destination. This field remain unchanged during the time the IPv4 datagram travel from source host to destination host.

4.12 Option

The Header of IPv4 data gram is made of two parts a fixed part and a variable part. The fixed part is 20 bytes long and variable part that can be maximum of 40 bytes.

There are 5 options

1. Strict source Routing
2. Loose source Routing
3. Record Routing
4. Time stamp
5. Padding



4.12.1 Strict Source Routing

A strict source routing is used by the source to predetermine a route for data gram as it travel through the internet.

4.12.2 Loose Source Routing

A loose source route option is similar to strict source route but it is less rigid. Each router in the list must visited, but the data gram can visit other router as well.

4.12.3 Record Routing

A record route option is used to record the internet routers that handle the data gram. It can list up to 9 router Address. All the Router are supposed to record their IP Address on their IP packets.

Note:

First 16 bits (2 byte) are reserved for option type (8 bit) and length (8 bit). Out of 40 bytes only 38 bytes are remaining for storing IPv4 addresses. In 38 bytes we can store 9 IPv4 addresses as each IPv4 address is of 4 byte

4.12.4 Time Stamp

It is used to find out delays at each router. Every router should record incoming time and outgoing time.

